

## Chapter: 19

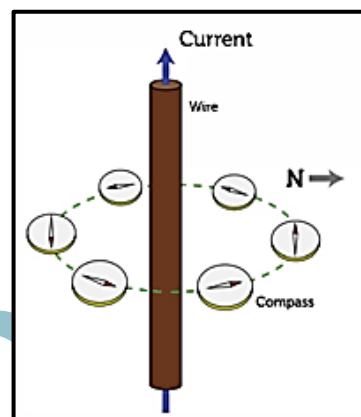
**ELECTROMAGNETIC INDUCTION****PRODUCTION OF ELECTRICITY BY MAGNETISM:****GENERATION OF ELECTROMAGNETIC FIELD:**

Since moving electric charges (i.e., an electric current) Produces a magnetic field around a current carrying conductor.

This magnetic field can be checked by putting a magnet compass near to current carrying wire, the needle of compass will be deflected, when it is placed near to current carrying conductor.

This shows that:

*Moving charges produce magnetic fields and in turn deflect magnetic compass.*



Such a demonstration for the first time was carried out in 1819 by Professor Hans Oersted of the University of Copenhagen.

**CONVERSION OF ELECTRIC ENERGY INTO MECHANICAL ENERGY:**

The working of electric motors is based on this simple phenomenon that:

A current carrying coil produces electromagnetic fields and when this coil is placed in a uniform magnetic field, due to the interaction of electromagnetic field around the conductors of coil and magnetic field of surrounding magnets, causing the coil to rotate.

In this case the electric energy is converted into mechanical energy. Mechanical energy can be used to do a variety of work, for example to pump water from ground level to a certain height.

**THE MOVING MAGNET PRODUCES ELECTRIC FORCE:**

To observe this phenomenon, connect a coil with a sensitive galvanometer and a magnet moves inside it.

*The relative motion of coil and magnet produces induced electric current.*

If the motion of magnet is stopped, induced current stops to exist.

The simple experiment (as shown in figure) is similar to the working of an electric generator.

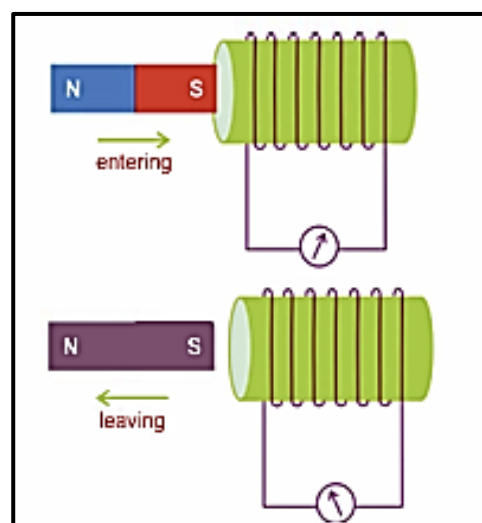
In the electric generator the conductor coil is rotated within the field of permanent magnet. The changing magnetic field produces induced electric current.

**Faraday's Experiment:**

In 1831 Micheal Faraday demonstrated that copper coils exposed to changing magnetic field produces electric current in the coil.

Michael Faraday demonstrated this phenomenon with the help of copper coil, a magnet and a galvanometer.

1. If the magnet was brought near to the coil an induced emf was produced and galvanometer showed current in one way.
2. When the magnet was moved from the coil away the direction of induced current was reversed, and galvanometer needle swung another way.



## ELECTROMAGNETIC INDUCTION:

Electromagnetic induction was discovered by the famous physicist Michael Faraday in the early 19th century. It explains the dynamic relationship between electricity and magnetism.

### Definition:

*The phenomenon in which an emf is induced in the coil due to the change in linking magnetic flux is called electromagnetic induction.*

### Explanation:

Joseph Henry and Faraday independently observed that an e.m.f is set up in a coil placed in a magnetic field whenever the flux through the coil changes. The effect is called electro-magnetic induction and if the coil forms a part of a closed circuit, the induced e.m.f causes a current to flow in the circuit.

The magnetic flux through a circuit can be changed in a number of different ways:

1. By changing the relative position of the coil with respect to a magnet or current carrying solenoid.
2. By changing current in the neighboring coil or by changing current in the coil itself.

## INDUCED ELECTROMOTIVE FORCE (EMF):

### Definition:

*"The current produced by moving the loop of wire across a magnetic field is known as induced current and e.m.f related to this current is called induced e.m.f."*

### WAYS OF PRODUCING INDUCED EMF:

Induced electromotive force (emf) can be generated in two primary ways:

1. By relative movement
2. By changing a magnetic field.

#### 1. BY RELATIVE MOVEMENT (THE GENERATOR EFFECT)

*"The induced emf produced due to the relative motion between a conductor and a magnetic field, such effect is called Generator Effect".*

This method is based on Faraday's Law of Electromagnetic Induction, which states that:

*"An emf is induced in a conductor when it experiences a change in magnetic flux".*

The generator effect, also known as motional emf, occurs when there is a relative motion between a conductor and a magnetic field.

**Example:** In a simple electrical generator, a coil of wire rotates within a magnetic field. This induced emf drives an electric current through an external circuit.

#### 2. BY CHANGING A MAGNETIC FIELD (THE TRANSFORMER EFFECT):

*"The induced emf produces due to a changing magnetic field around a stationary conductor, such effect is called Transformer Effect".*

This method also relies on Faraday's law but involves changing the strength or orientation of a magnetic field around a stationary conductor.

**Example:** In a transformer, an alternating current in the primary coil creates a changing magnetic field. This is the basis for how transformers transfer electrical energy between circuits at different voltages.

### CONCLUSION:

Both effects are fundamental principles in electromagnetism and are widely utilized in various electrical devices and systems.

## FARADAY'S LAW OF ELECTROMAGNETIC INDUCTION:

### Statement:

*The electromotive force (emf) induced in a closed circuit is directly proportional to the rate of change of magnetic flux passing through the circuit.*

### Explanation:

Faraday's law of electromagnetic induction is a fundamental principle in electromagnetism that describes the relationship between a changing magnetic field and the induced electromotive force (emf) or voltage in a closed circuit as illustrated in figure.

An e.m.f is induced in a coil through which the magnetic flux is changing. The magnitude of induced e.m.f depends only upon the number of turns and the time rate of change of flux linked with the circuit.

### Mathematically:

Consider a coil has "N" turns of wire be placed in the uniform magnetic field, if the magnetic flux through the coil is changed; an e.m.f is induced in the coil. The emf induced in the coil depends upon the rate of change of flux through coil as well as the number of turns of the wire in coil.

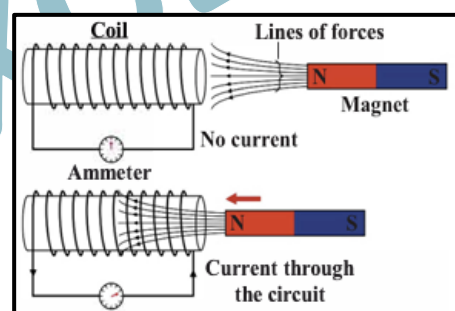
The induced e.m.f in the coil mathematically represented as:

$$\varepsilon = -N \frac{\Delta\phi}{\Delta t} \dots\dots\dots (1)$$

Where: "ε" = Induced electromotive force (emf) measured in volts (V).

$\frac{\Delta\phi}{\Delta t}$  = Rate of change of magnetic flux with respect to time, measured in weber per second (Wb/s or V).

N = Number of turns of the coil.



Negative sign represents that the induced emf opposes the change in flux, explained in Lenz's law.

### FACTORS AFFECTING THE MAGNITUDE OF THE INDUCED EMF:

The magnitude of the induced electromotive force (emf) in a circuit or coil is governed by several factors, which are primarily described by Faraday's law of electromagnetic induction. These factors include:

#### 1. Magnetic Flux Change:

If the rate of change of magnetic flux is higher, it will lead to a larger induced emf. Conversely, a slow change in the magnetic field will result in a smaller induced emf.

#### 2. Number of Turns in the Coil:

If a coil of wire has more turns, the greater will be the induced emf. Each turn of the coil contributes to the emf, so increasing the number of turns increases the overall emf.

#### 3. Area of the Coil:

The size of the coil or the area it encloses also affects the induced emf. A larger coil or a coil with a larger cross-sectional area will capture more magnetic flux lines, resulting in a larger induced emf.

#### 4. Angle between Magnetic Field and Coil:

The angle between the magnetic field lines and the plane of the coil affects the induced emf. When the magnetic field is perpendicular to the coil's plane (90°), the emf is maximized. When the field lines are parallel (0°), the emf is minimized.

#### 5. External Factors:

External factors, such as temperature, material property, pressure, and other environmental conditions, can also influence the induced emf, especially in situations where materials may exhibit nonlinear magnetic properties.

**SELF-ASSESSMENT QUESTIONS:**

1. Explain Faraday's law of electromagnetic induction and its significance in electrical systems.
2. How does the rate of change of magnetic flux affect the induced electromotive force (EMF) in a coil?

**LENZ'S LAW AND PRINCIPLE OF CONSERVATION OF ENERGY:****Introduction:**

After the introduction of Faraday's law of electromagnetic induction, Heinrich Friedrich Lenz formulated a rule for determining the direction of an induced current within a loop. According to Lenz's law

**Statement:**

*"The direction of induced current in a circuit is always such that it opposes the cause which produces it"*

**Explanation:**

This means that when a magnetic field through a loop of wire changes, it induces a current in the wire. The direction of this induced current is such that it creates its own magnetic field that *opposes* the original change.

*The induced current creates a magnetic field that opposes the original change in flux.*

**In Figure (a):** as the North Pole of the magnet approaches the surface of the coil, an induced current is generated within the coil in an anticlockwise direction.

Consequently, the coil generates a North magnetic pole, resulting in a repulsive force experienced by the approaching magnet. In this manner, the coil resists the factor responsible for inducing the electromotive force (emf).

**In Figure (b):** as the North Pole of the magnet moves away from the coil, an induced current is generated within the coil in a clockwise direction.

Consequently, the coil generates a South magnetic pole, resulting in an attractive force experienced by the direction of movement leaving magnet.

Hence the law states that:

*The induced current flows in a way that opposes the change which is giving rise to it.*

That is why negative sign is introduced in equation  $\varepsilon = -N \frac{\Delta\phi}{\Delta t}$

**LENZ'S LAW AND CONSERVATION OF ENERGY:**

The principle of conservation of energy states that energy cannot be created or destroyed; it can only change forms.

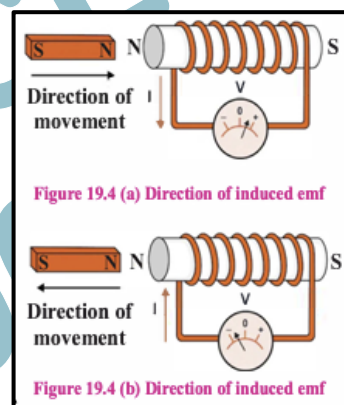
The opposing current requires work to be done against it (e.g., pushing a magnet into a coil), ensuring that energy input equals energy output.

When we apply this principle to electromagnetic induction, we see that the work done in changing the magnetic field is converted into electrical energy in the form of the induced current.

Hence Lenz's law is in accordance with the law of conservation of energy.

**SELF-ASSESSMENT QUESTIONS:**

1. Provide an example from everyday life or technology where Lenz's law is applied.
2. Discuss the consequences of not following Lenz's law in electromagnetic systems.



**WORKED EXAMPLES:**

- 19.1: A coil with 100 turns and an area of 0.02 square meters is placed in a changing magnetic field. The magnetic field changes from 0.2 Tesla to 0.5 Tesla in 0.1 seconds. Calculate the induced electromotive force (emf) in the coil, the direction of the induced current, and how this relates to Lenz's law.
- 19.2: A coil with 200 turns and an area of 0.03 square meters is placed in a uniform magnetic field of 0.4 Tesla. The coil is quickly pulled out of the magnetic field in 0.2 seconds. Calculate the induced electromotive force (emf) in the coil and determine the direction of the induced current.

**EDDY CURRENTS:****Definition:**

*Eddy currents are circulating currents induced in a conductor when it is exposed to a changing magnetic field.*

They are a common phenomenon in electromagnetic systems.

**Explanation:**

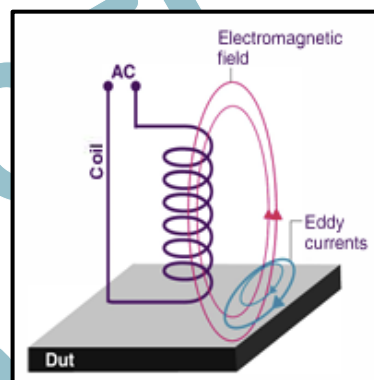
**Eddy currents** are loops of electric current induced within conductors by a changing magnetic field.

They swirl like little whirlpools, hence the name “eddy” and flow in closed paths perpendicular to the magnetic field.

Eddy currents are produced according to Faraday's law of electromagnetic induction, which states that:

*“A changing magnetic field induces an electromotive force (emf) in a conductor”.*

When a conductor (such as a metal plate or a coil of wire) is subjected to a changing magnetic field, the magnetic flux through the conductor changes, due to which eddy currents produce in the conductor (metal plate or coil) (as shown in figure). These currents can have both magnetic and heating effects.

**MAGNETIC EFFECTS OF EDDY CURRENTS:****1. COUNTERACTING MAGNETIC FIELD:**

Eddy currents generate their own magnetic fields, and the direction of these fields opposes the original magnetic field that induced them.

As a result, eddy currents create a magnetic field that counteracts the original magnetic field's change, thereby reducing the net magnetic field.

**2. MAGNETIC DAMPING:**

In applications like electromagnetic brakes and magnetic dampers, eddy currents are intentionally induced to create a magnetic resistance that opposes motion.

This magnetic damping effect is useful for controlling the movement of objects and slowing them down.

**HEATING EFFECTS OF EDDY CURRENTS:****JOULE HEATING:**

Eddy currents experience resistance as they flow through the conductor, and this resistance results in the conversion of electrical energy into heat, following Joule's law. Eddy currents can heat up the conductor.

In some cases, such as induction heating for metal processing or cookware, this heating effect is used for practical applications.



## **REDUCTION OF EDDY CURRENTS:**

In many electrical systems, eddy currents represent an undesirable source of energy loss, especially in transformers and electric motors.

To minimize these losses, laminated cores are used in transformers to break up the conducting paths and reduce the formation of eddy currents.

### **USE OF LAMINATED IRON CORES:**

The use of laminated iron cores in electric motors, generators, and transformers is essential for several important reasons:

In all three types of devices (electric motors, generators, and transformers), the iron cores are exposed to alternating magnetic fields. When a solid iron core is used, it can create circular electrical currents within the core material called eddy currents. These eddy currents dissipate energy in the form of heat and can be highly inefficient.

Laminated cores are constructed by stacking thin sheets or laminations of iron separated by insulating material. This design significantly reduces the formation of eddy currents because the laminations are electrically insulated from each other.

### **MINIMIZING ENERGY LOSSES DUE TO LAMINATING IRON CORE:**

Energy losses due to eddy currents are minimized by using laminated iron cores as follows.

#### **1. Enhanced Efficiency:**

By reducing the losses associated with eddy currents, laminated cores improve the overall efficiency of electric motors, generators, and transformers.

Less energy is wasted as heat, allowing these devices to operate more efficiently and with less energy consumption.

#### **2. Mitigation of Vibration and Noise:**

Eddy currents generated in a solid iron core can lead to vibrations and noise, which can be undesirable in many applications.

Laminated cores help reduce these vibrations and noise levels, making the devices quieter and more mechanically stable.

#### **3. Better Cooling and Thermal Management:**

Since laminated cores result in reduced heat generation due to decreased eddy current losses, they often allow for more efficient cooling and thermal management in these devices.

This can lead to longer operational lifetimes and improved reliability.

## **SELF-ASSESSMENT QUESTIONS:**

1. When might the heating effects of eddy currents be undesirable?
2. How do the dimensions of a conductor impact the formation and magnitude of eddy currents?

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## **SELF INDUCTION:**

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### **Definition:**

*The process, where a changing current in the coil induces an emf in the coil itself, is known as self-induction. OR*

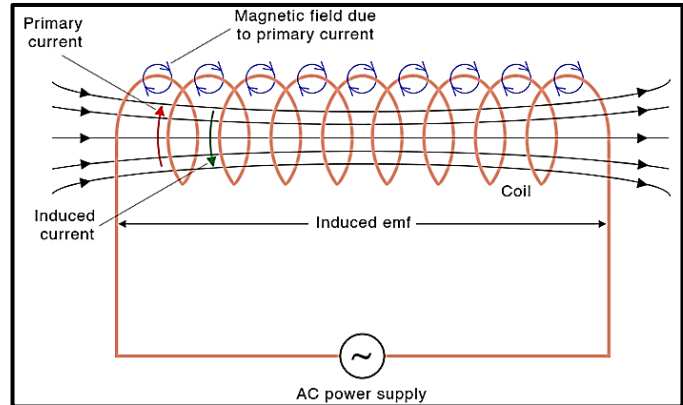
*The phenomenon in which an electric circuit opposes any change of current in itself is known as self-induction."*

**Explanation:**

When an electric current passes through a coil, it creates a magnetic field around it.

If the current in the coil changes (either increasing or decreasing), the magnetic flux within the coil also changes accordingly.

As a result of the change in magnetic flux, an induced emf is generated in the same coil, such an e.m.f is called self-induced e.m.f and the phenomena is known as self-induction.

**Mathematically:**

Consider the circuit (as shown in Figure), having a coil, connected in series with a battery and a rheostat.

When the rheostat is adjusted, it changes the current flowing through the circuit, leading to changes in the magnetic flux within the coil. These variations in magnetic flux result in the induction of an electromotive force (emf) within the coil.

If we denote the magnetic flux through a single loop of the coil as  $\phi$ , then the total magnetic flux passing through the  $N$  turns of the coil would be  $N\phi$ .

This relationship holds true because the magnetic flux, directly proportional to the magnetic field, which, in turn, is proportionate to the current ( $I$ ) flowing through the circuit

$$N\phi \propto I$$

$$N\phi = LI \dots\dots\dots (2)$$

Where  $L = \frac{N\phi}{I}$  is constant of proportionality called the self inductance of coil.

Self-inductance of the coil depends on factors such as the coil's number of turns, its cross-sectional area, and the material used for the core. Hence inductance of coil made up of soft iron is greater than the air core coil.

Equation (2) can be written as.  $N\Delta\phi = L \Delta I \dots\dots\dots (3)$

By the Faradays law of electromagnetic induction

$$\varepsilon = -N \frac{\Delta\phi}{\Delta t} \dots\dots\dots (4)$$

Substitute equation (3) in Equation (4), we get

$$\varepsilon = -L \frac{\Delta I}{\Delta t} \dots\dots\dots (5)$$

Equation (4) shows that the self-induced electromotive force (emf) within a coil is directly proportional to the rate of change of current within the coil.

In the field of electrical and electronics engineering, inductors find widespread use.

**Self-Inductance:**

We can define the self-inductance, denoted as "L," as:

*The ratio of emf to the rate of change of current in the coil is called Self Inductance.*

$$L = - \frac{\varepsilon}{\left(\frac{\Delta I}{\Delta t}\right)} \dots\dots\dots (6)$$

The negative sign in equation (6) signifies that the self-induced emf always opposes the cause that produces it. This is the reason why the self-induced emf is often referred to as the "back emf".

Consequently, due to the phenomenon of self-inductance, coils of wire are commonly known as "inductors".

**Unit of Inductance:**

The unit of self-inductance is the Henry (H), named after the American physicist Joseph Henry.

**One Henry (1 H)** is defined as:

*The amount of self-inductance in a circuit when a change in current of 1 ampere per second (1 A/s) induces an electromotive force (emf) of 1 volt (1 V) within the same circuit.*

Mathematically, this relationship can be expressed as:  $1 \text{ Henry} = \frac{1 \text{ Volt}}{1 \text{ Ampere per second}}$

## ENERGY STORED IN AN INDUCTOR:

An inductor is a passive electrical component that stores energy in the form of a magnetic field when an electric current flows through it.

### PRINCIPLE:

The ability of an inductor to store electric potential energy is based on the fundamental principle of electromagnetic induction.

### PROCESS OF ENERGY STORED IN AN INDUCTOR:

- 1. Current Flow:** When an electric current flows through a coil of wire (the inductor), it creates a magnetic field around the coil.
- 2. Magnetic Field Buildup:** As the current increases, the strength of the magnetic field around the coil also increases. This process takes a short amount of time, as the magnetic field does not build up instantly; it grows with the rate of change of the current.
- 3. Energy Storage:** The energy is stored in the magnetic field. The inductor stores electric potential energy in this magnetic field. The more current that flows through the inductor or the faster the current changes, the stronger the magnetic field, and thus, the more energy is stored.
- 4. Magnetic Field Collapse:** When the current decreases or stops (like when you turn off the power), the magnetic field starts to collapse. As it collapses, it induces an electromotive force (EMF) or voltage in the coil.
- 5. Released Energy:** This induced voltage represents the stored energy being released. The inductor converts the stored magnetic energy back into electric energy. This energy can be used to sustain the current for a short period or can be transferred to other parts of the circuit.

## ENERGY PRODUCED IN SELF INDUCTION:

An inductor stores energy in its magnetic field when it carries a direct current (DC). This energy remains stored as long as the inductor continues to carry the current.

But in alternating current (AC), when the current in the inductor increases, the stored energy also increases, and conversely, it decreases when the current is reduced.

### Expression for Energy produced in Self Induction:

Consider an inductor connected to a DC power source through a switch, as shown in Figure.

When the switch is closed, the current in the inductor gradually rises until it reaches its maximum value, denoted as  $I$ .

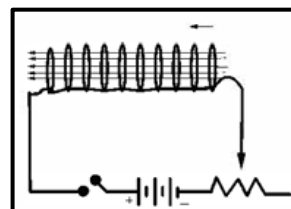
This changing current leads to a corresponding change in the magnetic flux within the coil, causing an induced electromotive force (emf) to establish itself in the coil.

Consequently, an induced current is generated in the circuit, which works to minimize the current produced by the battery in accordance with Lenz's law.

As a result, the battery must perform work on the charges to build up the current  $I$ . This work is mathematically expressed as follows:  $\Delta W = \Delta V \Delta q \dots\dots\dots (1)$

As the induced emf is produced in the inductor so.

$$\Delta V = \varepsilon_L = L \frac{\Delta I}{\Delta t}$$





$$\Delta W = E, \text{ energy stored}$$

And using work energy relation equation (1) can be written as.

$$E = \frac{\Delta I}{\Delta t} \cdot L \cdot \Delta q$$

Or  $E = \frac{\Delta q}{\Delta t} \cdot L \cdot \Delta I \dots\dots\dots (2)$

Since  $\frac{\Delta q}{\Delta t} = \text{Average Current}$

$$= \frac{0+I}{2}$$

$$= \frac{1}{2} I$$

And  $\Delta I = \text{Change in Current}$

$$= I - 0$$

$$= I$$

Therefore, equation (2) becomes.

$$E = \left(\frac{1}{2} I\right) L \cdot I$$

$$E = \frac{1}{2} LI^2$$

Where:  $E = \text{The energy stored in joules (J).}$

$L = \text{The self-inductance of the inductor in Henries (H).}$

$I = \text{The current flowing through the inductor in amperes (A).}$

### Conclusion:

This formula shows that:

*The energy stored in an inductor is directly proportional to the square of the current passing through it and is also dependent on the inductance of the inductor itself.*

As the current increases, the energy stored in the inductor increases, and as the current decreases, the stored energy decreases.

### SELF-ASSESSMENT QUESTIONS:

1. When is the energy stored in an inductor released?
2. Provide examples of practical applications where the energy stored in inductors is crucial.

### MUTUAL INDUCTANCE:

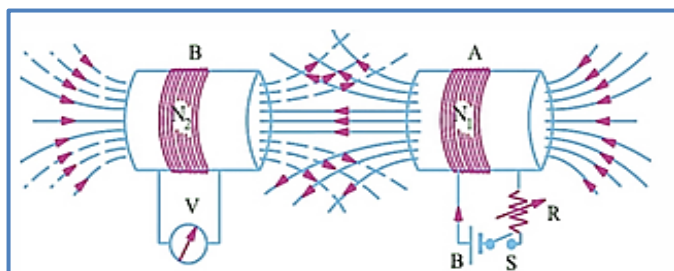
#### Definition:

*"A phenomenon where a changing current in one coil induces an electromotive force (EMF) in another coil is known as mutual induction".*

#### Explanation:

Imagine two closely positioned coils, as shown in Figure.

- The first coil, known as the primary coil, is connected to a battery through a variable resistor 'R' and a switch 'S.'
- The second coil, known as the secondary coil, is connected to a galvanometer.



When the resistance of primary coil by the rheostat changes, the current in the primary coil and the magnetic flux through the secondary coil also changes. By Faraday's law, an induced e.m.f produces in the secondary coil.

**Mathematically:**

Consider the two closely wound coils of wire (as shown in Figure). The current  $I_p$  in primary coil A, which has  $N_p$  turns, creates a magnetic field.

Some of the magnetic field lines pass through secondary coil B, which has  $N_s$  turns.

Let the flux passing through secondary coil having  $N_s$  turns be is  $N_s\phi_s$ . This magnetic flux depends on the primary coil current  $I_p$ .

$$N_s\phi_s \propto I_p$$

$$N_s\phi_s = M I_p$$

The change of current in primary coil changes the magnetic flux in secondary coil. Therefore

$$N_s \cdot \Delta\phi_s = M \Delta I_p \dots\dots\dots (1)$$

Here, “M” represents the proportionality constant known as the mutual inductance between the two coils.

Mutual inductance depends on factors such as:

- (1) The number of turns in the coils.
- (2) Their geometrical characteristics.
- (3) The materials are used to construct the coils.
- (4) The separation between them.

When the separation between the coils increases, the mutual inductance decreases because of the diminishing magnetic flux.

From Faradays’ law of electromagnetic induction

$$\varepsilon_s = -N_s \frac{\Delta\phi_s}{\Delta t} \dots\dots\dots (2)$$

Substitute Equation (1) in Equation (2), we get

$$\varepsilon_s = -M \frac{\Delta I_p}{\Delta t} \dots\dots\dots (3)$$

Equation (3) shows that induced emf produced in the secondary coil is equal to negative rate of change of current in the primary coil and this equation can be used to provide the definition of mutual inductance.

$$M = - \frac{\varepsilon_s}{\left(\frac{\Delta I_p}{\Delta t}\right)} \dots\dots\dots (4)$$

Here, proportionality constant “M” is Mutual inductance, which measures the ability of one coil to induce an electromotive force (EMF) in another coil when the current in the first coil changes.

**Unit of Mutual Inductance:**

The unit of Mutual inductance is Henry. One Henry (H) of mutual inductance can be defined as:

*“The mutual inductance between two coils is one Henry, when an EMF of one volt is induced in the secondary coil when the rate of change of current in the primary coil is one ampere per second”.*

$$1\text{Henry} = \frac{1\text{Volt}}{1\text{ampere/sec ond}}$$

**SELF-ASSESSMENT QUESTIONS:**

1. Give an example of devices or systems where mutual induction is utilized
2. What are the advantages of using mutual induction in practical applications.

**TRANSFORMER:****Definition:**

*A transformer is an electrical device used to transfer electrical energy between two or more coils of wire through electromagnetic induction.*

**Principle of Working:**

It works on the principle of mutual induction.

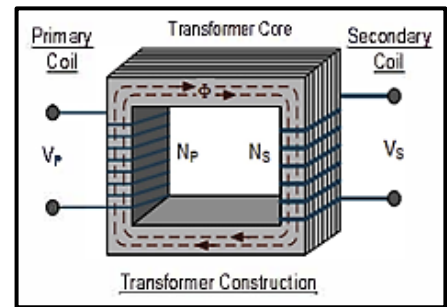
**CONSTRUCTION OF A TRANSFORMER:**

The transformer consists of the following main components as shown in the figure.

**1. Core:**

Transformers consist of two coils of wire, known as the primary coil and the secondary coil, wound around a common magnetic core.

The core is typically made of materials with high magnetic permeability, such as laminated iron or ferrite, to enhance the efficiency of the transformer.

**2. Primary Coil:**

The primary coil is connected to the input voltage  $V_p$  source. It consists of a specific number of turns of wire wound around one section of the core denoted as  $N_p$ .

**3. Secondary Coil:**

**Secondary Coil** The secondary coil is connected to the load or the device that needs the transformed voltage. It has a different number of turns of wire wound around another section of the core denoted by  $N_s$ .

**4. Insulation:**

Both the primary and secondary coils are insulated from each other to prevent electrical contact and ensure electrical isolation.

**WORKING OF TRANSFORMER:**

Imagine we have an alternating electromotive force (emf) applied to the primary coil. If, at a certain moment (t), the magnetic flux within the primary coil is changing at a rate of  $\frac{\Delta\phi}{\Delta t}$ , this change in flux will induce a counteracting electromotive force (emf) in the primary coil, opposing the applied voltage.

We can express the instantaneous value of this self-induced emf as follows:

$$\text{Self-induced emf, } V = -N_p \frac{\Delta\phi}{\Delta t}$$

If the coil's resistance can be considered negligible, then the opposing emf in the primary coil is equal in magnitude but opposite in direction to the applied voltage.

This relationship can be represented as:

$$V_p = -N_p \frac{\Delta\phi}{\Delta t} \dots\dots\dots (1)$$

Here,  $N_p$  represents the number of turns in the primary coil.

Now, let's assume that the flux passing through the primary coil also flows through the secondary coil. In other words, these two coils are magnetically linked.

Hence, the rate of change of magnetic flux in the secondary coil will also be  $\frac{\Delta\phi}{\Delta t}$  and the magnitude of the induced emf across the secondary coil can be expressed as:

$$V_s = -N_s \frac{\Delta\phi}{\Delta t} \dots\dots\dots (2)$$

Where  $N_s$  is number of turns in secondary. When we divide equation (1) by equation (2), we obtain

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

**TYPES OF TRANSFORMERS:**

There are two types of Transformers.

1. Step up transformer

2. Step-Down Transformers:

Transformers can either step up (increase) or step down (decrease) the voltage of an alternating current while keeping the frequency of the alternating current unchanged.

The ratio of the number of turns in the primary coil ( $N_p$ ) to the number of turns in the secondary coil ( $N_s$ ) determines whether the transformer is a step-up or step-down transformer.

### **1. STEP UP TRANSFORMER:**

*If  $N_s > N_p$ , then it's a step-up transformer and it increases the voltage.*

#### **CHARACTERISTICS OF STEP-UP TRANSFORMER:**

##### ***1. Increasing Voltage:***

Step-up transformers are designed to increase the voltage of electricity.

They have more turns in the secondary coil than in the primary coil, resulting in a higher secondary voltage compared to the primary voltage.

##### ***2. Decreasing Current:***

Increasing the voltage through a step-up transformer reduces the current while maintaining the same power ( $P = VI$ ). Lower current reduces the resistive losses in the transmission cables.

The power lost as heat (FR losses) is proportional to the square of the current, so decreasing the current significantly reduces these losses.

##### ***3. Efficient Long-Distance Transmission:***

High voltage is essential for efficient long-distance power transmission.

Step-up transformers are used at power generation plants to raise the voltage, allowing electricity to be transported over extensive electrical grids with minimal energy loss. This is especially important for transmission lines covering large distances.

### **2. STEP-DOWN TRANSFORMERS:**

*If  $N_p > N_s$ , it is a step-down transformer and it decreases the voltage.*

#### **CHARACTERISTICS OF STEP-DOWN TRANSFORMER:**

##### ***1. Decreasing Voltage:***

Step-down transformers, on the other hand, lower the voltage from high levels to safer, more usable levels for homes and industries. They have fewer turns in the secondary coil, which decreases the secondary voltage.

##### ***2. Increasing Current:***

While step-down transformers decrease voltage, they increase the current to maintain the same power.

##### ***3. Beneficial Local Distribution:***

This is beneficial for local distribution because it ensures that the power can be delivered to homes, businesses, and industrial facilities while minimizing FR losses over shorter distances.

### **SELF-ASSESSMENT QUESTIONS:**

1. Differentiate between a step-up transformer and a step-down transformer, providing examples of each.
2. Name the fundamental principle behind the operation of a transformer

### **TRANSMISSION OF ELECTRICITY:**

Step-down and step-up transformers are used for the electric supply from power stations to houses and electric appliances. Here's how they are used in the electric supply process as shown in figure.

**1. POWER GENERATION AT THE STATION:**

Electricity is generated at power stations, often using generators powered by various sources such as hydel, coal, natural gas, nuclear energy, or renewable sources like wind and solar. The electricity generated is typically produced at a relatively low voltage level.

For efficient generation and long-distance transmission, higher voltages are preferred to minimize energy losses.

**2. STEP-UP TRANSFORMERS AT THE POWER STATION:**

To raise the voltage to levels suitable for long-distance transmission, step-up transformers are employed at the power station.

Step-up transformers increase the voltage and reduce the current, which lowers the  $I^2R$  losses, making electricity transmission more efficient.

High-voltage transmission lines, such as overhead power lines or underground cables, carry electricity over long distances to substations.

**3. TRANSMISSION AND SUBSTATIONS:**

The high-voltage electricity is transmitted over transmission lines to substations located at various points in the electricity distribution network.

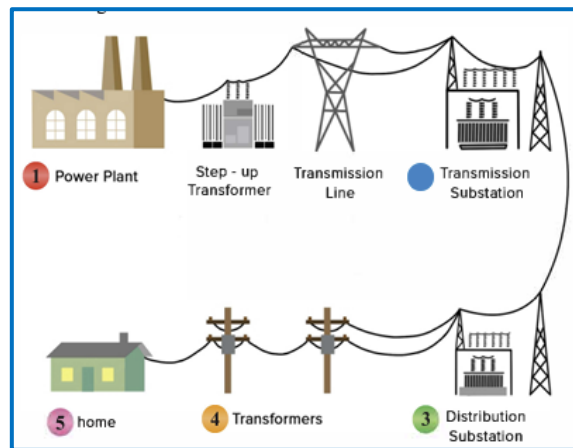
At these substations, step down transformers are used to lower the voltage to a more manageable level for further distribution.

**4. STEP-DOWN TRANSFORMERS IN SUBSTATIONS:**

Step-down transformers reduce the high-voltage electricity to lower, safer voltage levels, suitable for local distribution. This lower voltage is then distributed to houses through distribution lines.

**WORKED EXAMPLES:**

- 19.3: Suppose a step-up transformer has 200 turns in the primary coil and 800 turns in the secondary coil. If the voltage in the primary coils is 120 volts, calculate the voltage in the secondary coil.
- 19.4: Let a step-down transformer have 600 turns in the primary coil and 150 turns in the secondary coil. The voltage in the primary coil is 240 volts. Calculate the voltage in the secondary coil.

**MOTIONAL EMF:****Definition:**

*Motional electromotive force (emf) is a phenomenon that arises when a conductor moves through a magnetic field, inducing an electromotive force within the conductor.*

**Explanation:**

This concept is based on Faraday's law of electromagnetic induction and is a fundamental aspect of electromagnetism.

When a coil of wire rotates in a magnetic field, motional emf is induced, leading to the generation of electric current. The Motional electromotive force (emf) is the voltage induced in a conductive rod as it moves through a magnetic field.

**Applications:**

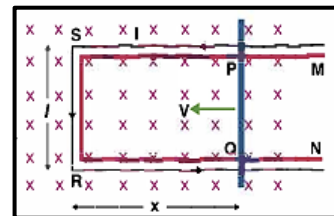
Motional emf commonly appears in various practical applications. For example, the generation of electricity by a generator.



### Expression for the motional emf

Consider a straight wire PQ having length “L” as shown in figure. The conductor is in motion within a rectangular loop PQRS. This motion occurs within a uniform magnetic field B, perpendicular to the plane of the page.

We assume that the rod moves uniformly at a constant velocity of “v” meters per second, and the surface it moves on is frictionless.



Each free electron of the conductor is moving with the conductor and thus experiences a force:

$$\vec{F} = q (\vec{v} \times \vec{B}) \dots\dots\dots (1) \quad (q = e)$$

And the magnitude of force is  $F = q v B \sin\theta$

Electrons continue to accumulate at one end of the conductor, creating an excess of electrons, which results in a negative charge at that end. Conversely, the opposite end, with a deficiency of electrons, leads to the development of a positive charge. This process generates a potential difference within the wire, which is termed as motional electromotive force (emf).

In order to derive an expression for the motional emf, we use the definition of potential difference.

Since the potential difference is defined as work done per unit charge

$$V = \frac{W}{q} \dots\dots\dots (2)$$

As  $W = Fd \cos\theta$

Where d = L

Then  $W = FL \cos 180^\circ$

Angle between F and L is  $180^\circ$

$$W = FL (-1)$$

$$W = -FL$$

Put the value of W in equation (2), we get

$$V = \frac{-FL}{q}$$

By putting the value of “F”, we get

$$V = \frac{-(q v B \sin\theta)L}{q}$$

$$= -v B L \sin\theta$$

If the conductor is moving with right angle to the magnetic field, then  $\theta = 90^\circ$

$$V = -v B L \dots\dots\dots (2)$$

Where “V” is motional emf, “v” represents the velocity of conductor moving in a magnetic field and “L” is the length of conductor

### SELF-ASSESSMENT QUESTIONS:

1. Describe Fleming's right-hand rule and its application in the context of motional EMF.
2. How does the direction of motion, magnetic field, and current relate in Fleming's righthand rule?

### WORKED EXAMPLE:

19.5: A straight conductor of length  $L = 0.5$  meter moves at a velocity of  $2$  m/s perpendicular to a magnetic field  $B = 0.1$  Tesla. Calculate the motional emf induced in the conductor.

## AC GENERATOR:

### Definition:

*An AC generator, also known as an alternator, is a device that converts mechanical energy into electrical energy in the form of alternating current (AC).*

### Principle of Working:

It's based on the principle of electromagnetic induction, which was first described by Michael Faraday.

**CONSTRUCTION OF AN AC GENERATOR:**

An AC generator typically consists of the following four components as shown in figure:

- (1) Armature                      (2) Field magnet                      (3) Sliprings                      (4) Brushes

**1. Armature:**

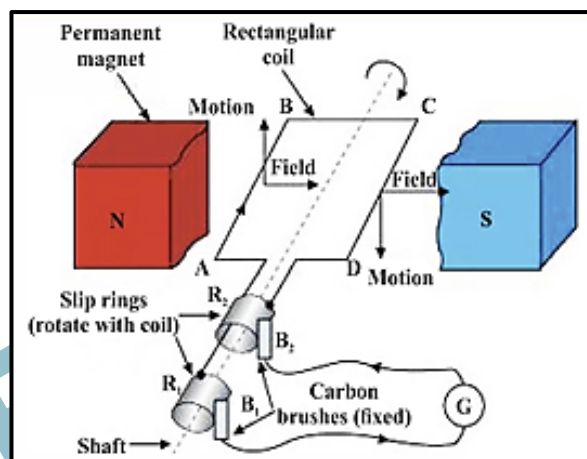
- The armature or rotor is a rectangular coil mounted on a rotating shaft.
- It is a component that spins within a magnetic field to induce electrical current.

**2. Permanent magnet:**

- The permanent magnet or stator is a stationary part of the generator that produces a magnetic field.
- It is typically made up of a set of magnets or electromagnets arranged in a circular or cylindrical configuration around the rotor.

**3. Slip Rings and Brushes:**

- The ends of the rotor coil are connected to slip rings, which are conductive rings that rotate with the rotor.
- Brushes, typically made of graphite, press against the slip rings to collect the generated electrical current.

**4. Shaft and Bearings:**

- The rotor is mounted on a shaft that allows it to rotate freely.
- Bearings are used to reduce friction and enable smooth rotation.

**WORKING OF AN AC GENERATOR:**

The working of an AC generator involves several steps:

**1. Rotation of the Armature coil:**

The coil or rotor is mechanically rotated using an external source of power, such as an engine, a turbine, or any other energy source.

**2. Generation of a Changing Magnetic Field:**

As the rotor spins within the stator's magnetic field, the magnetic field within the rotor coil changes. This changing magnetic field induces an electromotive force (emf) or voltage in the coil,

**3. Generation of Alternating Current:**

The induced voltage causes an electric current to flow through the coil, and since the magnetic field's direction is changing, the current produced is alternating in nature.

This means the direction of the current constantly switches back and forth, resulting in an AC output.

**4. Collection of Output:**

The alternating current generated in the rotor coil is collected using slip rings and brushes. The brushes maintain contact with the slip rings as they rotate, allowing the generated AC to be drawn from the generator.

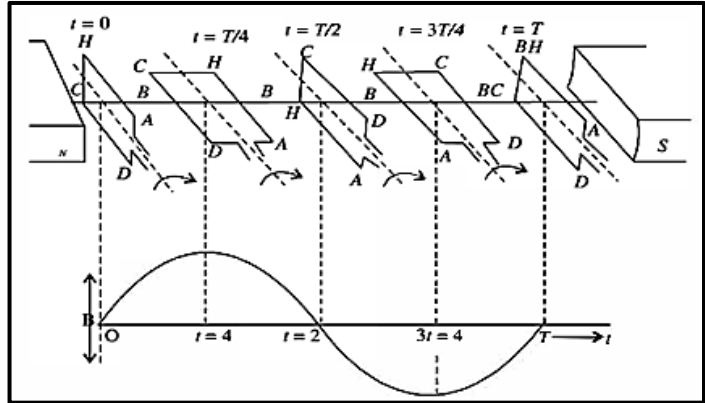
**EXPLANATION OF THE WORKING OF AC GENERATOR:**

Suppose the armature coil AHCD rotates counterclockwise. As it rotates, the magnetic flux linked with it changes, inducing a current in the coil. The direction of the induced current follows Fleming's right-hand rule.

### During 1<sup>st</sup> Half Rotation:

At first, the plane of the coil is perpendicular to the field. At this moment the side AB and CD are moving parallel to the field so the induced e.m.f is zero.

When the armature is in the horizontal position and rotates clockwise, wire AH moves downward, while DC moves upward. This causes the induced electromotive force (emf) to flow from H to A and from D to C, within the coil, forming a current path of DCHA.



During the 1<sup>st</sup> quarter of rotation of AH, induced e.m.f reaches from zero to maximum value and in next quarter of rotation it decreases to zero.

Therefore, the direction of the induced emf and the current alternates after every half revolution in the external circuit as well.

Consequently, the current produced switches direction in each cycle.

### During 2<sup>nd</sup> Half rotation:

During the 2<sup>nd</sup> half, wire AH moves upward, and wires CD move downward, so the direction of the induced e.m.f is also changed.

But the variation of its magnitude is the same as in the first half, leading to a reversal in the direction of induced current within the coil to AHCD.

*The current caused by such an e.m.f in any circuit is called alternating current.*

### EXPRESSION FOR INDUCED EMF:

When the coil is rotated, a motional electromotive force (emf) is generated on each of its sides, but these emfs have opposite directions because the two sides are moving in opposite directions relative to the magnetic field. However, the other two sides of the coil are moving in the same direction with respect to the field. Consequently, emfs are induced in these sides in the same direction, leading to a mutual cancellation effect. As a result, the total emf induced in the coil is given by

$$E = -2v B N l \sin(\omega t) \dots\dots\dots (1)$$

Where  $v$  = linear velocity,  $B$  = magnetic field,  
 $N$  = Number of turns,  $l$  = length of coil  $w$  = angular velocity.

Since each particle of the sides AH and CD rotate in a circle of radius equal to the width of the coil  $b$ . Therefore

$$v = r \times \omega$$

$$= b/2 \times \omega$$

By putting in equation (1), we get

$$E = -2(b/2 \times \omega) B N l \sin(\omega t)$$

$$E = -\omega(l \times b) B N \sin(\omega t) \dots\dots\dots (2)$$

Since  $b \times l = A$ ; Area. Therefore, equation (2) becomes

$$E = -\omega A B N \sin(\omega t) \dots\dots\dots (3)$$

When,  $\theta = 90^\circ$ , Induced e.m.f will be maximum which is denoted by  $E_0$ . So

$$\text{Maximum e.m.f; } E_0 = -BNA\omega$$

Hence, we can write equation (3) as follows:

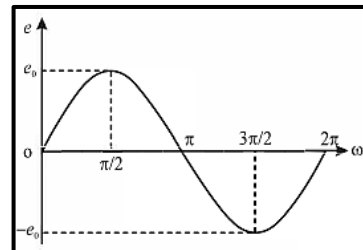
$$\text{Induced e.m.f; } E = E_0 \sin\theta$$

Where  $E_0 = BNA\omega$  is the maximum or peak value of the emf which depends on the area and number of the coil, intensity of the field and speed of rotation.

The expression for the generated emf can also be expressed in terms of the frequency “f” (i.e. number of turns per second).

$$E = E_0 \sin 2\pi ft$$

Figure (b) shows the variation of emf versus ( $\omega$ ). It shows that the direction of the emf is reversed after every half revolution of the coil.



### SELF-ASSESSMENT QUESTIONS:

1. How does a rotating magnetic field contribute to the rotation of the motor's rotor?
2. Discuss common methods used for starting AC motors.
3. Provide examples of applications where AC motors are commonly used.

### WORKED EXAMPLE:

19.6: Find the frequency of an AC generator if the voltage output is given as  $E_0 = 200$  V.

### A.C MOTOR:

#### Definition:

*An AC (Alternating Current) motor is a device designed to convert electrical energy into mechanical energy by using alternating current.*

### CONSTRUCTION OF A.C MOTOR:

#### PRIMARY COMPONENTS OF AC MOTOR:

There are various types of AC motors, but the main features and components are generally consistent across different designs.

The primary components as shown in figure and their roles in an AC motor are as follows:

#### 1. Stator:

- The stator is the stationary part of the motor and contains the primary windings.
- Its main role is to produce a rotating magnetic field when AC voltage is applied.
- The rotating magnetic field interacts with the rotor to induce motion.

#### 2. Rotor:

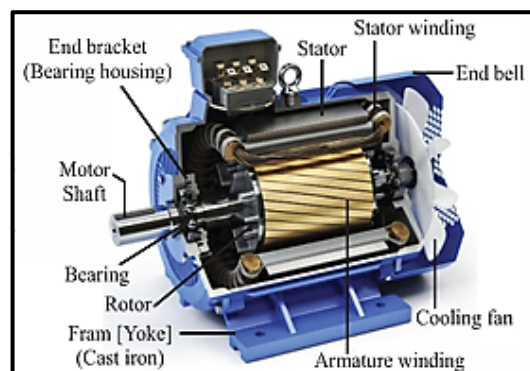
- The rotor is the rotating part of the motor.
- It can be of different types, such as squirrel cage or wound rotor, depending on the motor design.
- When the stator produces a rotating magnetic field, the rotor experiences a torque due to interaction with the field. This torque causes the rotor to rotate and generate mechanical output.

#### 3. Bearings:

- Bearings are essential components that support and allow the rotor to rotate within the stator.
- They reduce friction, enabling smooth and efficient operation of the motor.

#### 4. Shaft:

- The shaft is connected to the rotor and extends beyond the motor housing.



- It transfers mechanical energy to the outside world, allowing the motor to perform useful work when connected to a load.

### 5. Cooling System:

- Many AC motors incorporate cooling systems, such as fans or fins, to dissipate heat generated during operation.
- Efficient cooling is important to prevent overheating and prolong the motor's lifespan.

## WORKING OF AC MOTOR:

The main features and components work together to enable the AC motor to function.

- When AC voltage is applied to the stator windings, a rotating magnetic field is created, which exerts a torque on the rotor.
- The rotor's rotation results in mechanical work being performed by the motor, and the motor can be used to drive various mechanical devices, such as fans, pumps, conveyor belts, and more.

## **EFFICIENCY & PERFORMANCE OF AC MOTOR:**

The efficiency and performance of an AC motor depend on the design and quality of its components, the load it drives, and its operating conditions.

Proper maintenance and control mechanisms, such as speed control, are often used to optimize the motor's performance in different applications.

## THE PRODUCTION OF BACK EMF IN ELECTRIC MOTORS:

The concept of back electromotive force (back EMF) in electric motors is fundamental to understanding motor operation and efficiency.

### **BACK EMF:**

*Back EMF is a self-generated electromotive force that opposes the applied voltage in a motor.*

It plays a crucial role in motor behavior, especially in limiting current and regulating speed.

### **BACK EMF IN ELECTRIC MOTORS:**

Here's a detailed explanation of the concept of back EMF in electric motors:

#### 1. Electromagnetic Induction:

Back EMF is a consequence of Faraday's law of electromagnetic induction, which states that a change in magnetic flux through a coil of wire induces an electromotive force (EMF) in that coil.

In an electric motor, the coil of wire is typically part of the rotor or armature.

#### 2. Rotating Magnetic Field:

When an electric motor is powered, the stator (the stationary part of the motor) generates a rotating magnetic field by applying an alternating current (AC) voltage to its windings.

In a direct current (DC) motor, the commutator and brushes create a changing magnetic field.

#### 3. Rotor or Armature Interaction:

The rotor (or armature) in the motor is mounted within the magnetic field created by the stator.

As the rotor rotates, it cuts through the magnetic field lines, causing a change in magnetic flux within the coils of wire on the rotor.

#### 4. Back EMF Generation:

The change in magnetic flux through the rotor windings induces a voltage, which is referred to as back EMF. The direction of the induced voltage is in opposition to the applied voltage.



In other words, it generates a voltage that resists the flow of current through the motor windings.

### 5. Current Regulation:

Back EMF has a critical function in motor operation. As the motor speeds up, the back EMF increases. This increase in back EMF results in a decrease in the net voltage across the motor windings.

Consequently, the current flowing through the motor decreases. This self-regulation mechanism is essential for preventing the motor from drawing excessive current, overheating, and potentially damaging itself.

### 6. Effect on Motor Speed:

The relationship between back EMF and motor speed is inverse. When the motor operates at a higher speed, the back EMF is greater, and the current is reduced, helping maintain a consistent speed.

Conversely, at lower speeds or when subjected to a higher mechanical load, the back EMF decreases, allowing more current to flow and providing the necessary torque to overcome the load.

### CONCLUSION:

In summary, the concept of back EMF in electric motors is the self-induced voltage that opposes the applied voltage and helps regulate the current and speed of the motor.

It is a critical factor in ensuring the safe and efficient operation of electric motors, and understanding it is essential for motor design, control, and performance analysis.

### SELF-ASSESSMENT QUESTIONS:

1. Discuss how the production of back EMF aligns with the principle of energy conservation.
2. What is the primary purpose of the back EMF in electric motors?

### SUMMARY

- When a current flows through the conductor magnetic field is produced around it.
- The relative motion of coil or magnet will produce induced emf. If the motion of magnet and coil is stopped induced current ceases to exist.
- The magnitude of the induced emf depends on the change of magnetic flux, number of turns in the coil, area of the coil, angle between magnetic field and coil.
- When the magnetic flux  $\phi_B$  passing through a closed conducting loop changes over time, it results in the generation of a current and an electromotive force (emf) within the loop.
- Lenz's law states that "The direction of induced emf in a circuit is always such that it opposes the cause which produces it".
- Lenz's law is in accordance with the law of conservation of energy.
- If a current  $I$  in a coil change with time, an emf is induced in the coil. This self-induced emf is given by
$$\varepsilon = -L \Delta I / \Delta t$$
- An inductor stores energy in the form of a magnetic field when current flows through it, and this stored energy is proportional to the square of the current and the inductance of the inductor.
- If primary and secondary coils are near each other, a changing current in either coil can induce an emf in the other is known as mutual inductance.
- A transformer is an electrical device that transfers energy between two or more circuits using electromagnetic induction. It changes voltage levels while keeping power conservation and efficiency intact. The transformer has two or more coils, called windings, connected by a magnetic core, which allows it to increase (step-up) or decrease (step-down) voltage.
- Motional emf is the electromotive force induced in a conductor moving through a magnetic field, resulting from the relative motion between the conductor and the magnetic field lines.
- An AC generator converts mechanical energy into electrical energy in the form of alternating current.

- An AC motor converts electrical energy into mechanical energy.

## **TEXTBOOK WORK**

### **SECTION- A: MULTIPLE CHOICE QUESTIONS (MCQS)**

- Electricity production by magnetism involves
  - (a) The conversion of chemical energy to electrical energy.
  - (b) The use of static electricity for power generation.
  - (c) The movement of conductors within magnetic fields.
  - (d) The absorption of solar energy by photovoltaic cells.
- Induced e.m.f can be generated by relative movement, known as the generator effect, and by changing magnetic fields, known as the transformer effect.
  - (a) True in transformers only.
  - (b) False in all cases.
  - (c) Applicable to both cases.
  - (d) True only for static magnetic fields.
- The magnitude of induced e.m.f. increases with
  - (a) The decrease in the speed of the conductor's motion.
  - (b) The decrease in the strength of the magnetic field.
  - (c) The increase in the number of turns in the coil.
  - (d) The reduction in the area of the coil.
- Faraday's law of electromagnetic induction states
  - (a) That current in a circuit always opposes the change in magnetic flux.
  - (b) That the induced e.m.f. is directly proportional to the rate of change of magnetic flux.
  - (c) That a changing electric field produces a magnetic field.
  - (d) magnetic fields can only be produced by electric currents.
- Lenz's law predicts the direction of an induced current (2024)
  - (a) To conserve electric charge.
  - (b) By stating that the induced current will oppose the change causing it.
  - (c) By following the direction of the applied magnetic field.
  - (d) Based on the magnitude of the electric field.
- Eddy currents produce:
  - (a) Static electricity in conductive materials.
  - (b) Alternating magnetic fields with no heating effects.
  - (c) Magnetic fields that oppose the inducing field and cause heating in conductors.
  - (d) Uniform electric fields that have no impact on power generation.
- Laminated iron cores are used in electric motors and transformers to
  - (a) Reduce the weight of the device.
  - (b) Enhance the mechanical strength.
  - (c) Minimize eddy current losses and reduce heating.
  - (d) Increase the magnetic permeability of the core.
- Self-induction is the phenomenon where:
  - (a) A changing electric field induces a magnetic field.
  - (b) A changing magnetic field within a coil induces an emf in the same coil.
  - (c) A constant magnetic field induces a constant emf.
  - (d) Two coils induce emf in each other through mutual induction.

9. An inductor stores electric potential energy in:
- (a) The electric field around it.
  - (b) The magnetic field within its coil.
  - (c) The capacitance of its windings.
  - (d) The heat generated by its resistance.
10. Transformers work on the principle of
- (a) Converting direct current (DC) to alternating current (AC).
  - (b) Electromagnetic induction between primary and secondary coils.
  - (c) Generating electricity through chemical reactions.
  - (d) Using permanent magnets to maintain a constant voltage.

### **SECTION- B: SHORT ANSWERED QUESTIONS**

1. Explain Faraday's law of electromagnetic induction.
2. State Lenz's law and discuss its significance in the context of electromagnetic induction.
3. How does a step-up transformer differ from a step-down transformer?
4. Describe the principle of operation of an AC generator. (2024)
5. Define self-induction and explain how it occurs in a coil.
6. Define mutual induction and provide an example of a system exhibiting mutual induction.
7. How does the arrangement of coils influence the degree of mutual induction between them?
8. Define motional electromotive force (emf)?
9. Differentiate between motional emf and electromagnetic induction in terms of their fundamental principles.
10. Differentiate between AC (alternating current) and DC (direct current) in the context of long-distance power transmission?

### **SECTION- C: LONG ANSWERED QUESTIONS**

1. How does the strength of the magnetic field affect the induced electromotive force (emf) in a coil according to Faraday's law? b. Explain the relationship between the magnetic field strength and the magnitude of the induced emf.
2. In Faraday's law, why is the area of the coil an important factor in determining the induced emf?
3. How does changing the orientation of a coil with respect to the magnetic field influence the induced emf? Also describe the role of the number of turns in a coil in the context of induced emf. How does increasing the number of turns in a coil affect the induced emf, and why?
4. According to Faraday's law, why is the rate of change of magnetic flux crucial for inducing emf? Discuss how the speed of motion or change in magnetic field influences the induced emf.
5. How does the resistance of the conductor impact the flow of induced current in a circuit? In what situations might the resistance of the conductor become a significant factor in induced emf?
6. Explain the relationship between the frequency of the changing magnetic field and the induced emf. How does the frequency of the AC source affect the generation of induced emf in a coil?
7. Discuss how the material properties of a coil, such as its conductivity, can influence induced emf. In what ways might the type of material use in a coil affect its response to changing magnetic fields?
8. Why is the presence of a core material significant in transformers concerning induced emf? Explain how the type and properties of the core material can impact the efficiency of a transformer.
9. Why are slip rings used in AC generators, and how do they differ from commutator used in DC generators.
10. Why a commutator is necessary in DC motors, and how does it facilitate the continuous rotation of the motor?
11. What are the common applications of DC motors in everyday devices?

**SECTION- D: NUMERICAL**

- Two coils are placed adjacent to each other, and a change in current in the first coil induces an emf of 0.5 V in the second coil. If the mutual inductance is 0.2 H, calculate the rate of change of current in the first coil. **(2.5A/s)**
- A coil with an inductance of 0.5 H experiences a rate of change of current of 2 A/s. Calculate the induced EMF in the coil. **(-1 V)**
- Two coils are placed close to each other. If a change in current of 3 A/s in the first coil induces an EMF of 4 V in the second coil, calculate the mutual inductance. **(1.33H)**
- A transformer has 200 turns in the primary coil and 400 turns in the secondary coil. If the primary voltage is 120 V, calculate the secondary voltage for a step-up transformer. **(240V)**
- An inductor with an inductance of 0.02H has a current flowing through it of 2 A. Calculate the energy stored in the inductor. **(0.04J)**
- A coil stores energy in form of electric potential energy of 0.2J when it carries a current of 2A. Calculate the inductance of coil. **(0.1H)**
- An AC generator produces an alternating current with a maximum voltage of 240 V. If the frequency of the generated AC is 50 Hz, calculate the peak value of the voltage. **(339V)**
- A transformer has 1000 turns on its primary coil and 200 turns on its secondary coil. If the primary voltage is 120 V, calculate the secondary voltage for a step-down transformer. **(24V)**
- A conductor of 0.4 m moves at a velocity of 15 m/s perpendicular to a magnetic field of 0.3 T. Calculate the motional EMF induced in the conductor. **(0.6V)** **(MP:25)**

**ASSIGNMENT: 19****SECTION- A: MULTIPLE CHOICE QUESTIONS****Faraday's Laws, Lenz Law, Motional e.m.f:**

- Lenz's law predicts the direction of an induced current **(2024)**
  - to conserve electric charge
  - by stating that the induced current will oppose the change causing it
  - by following the direction of the applied magnetic field
  - based on the magnitude of the electric field
- Lenz's law is direct consequence of this: **(2022)**
  - Ohm's Law.
  - Coulomb's law
  - Faraday's law
  - law of conservation of energy**
- Induced emf cannot be produced by: **(2022)**
  - change of flux
  - movement of a coil
  - AC source
  - DC source**
- If the speed of a conductor moving in a uniform magnetic field perpendicularly is increased, the induced emf: **(2017)**
  - Increases**
  - decreases
  - remains constant
  - becomes zero
- The direction of induced current is given by: **(2013, 05)**
  - Ohm's law
  - Lenz's law**
  - Coulomb's law
  - Ampere's law
- The motional e.m.f. induces in a coil is independent of: **(2010)**
  - Change of flux.
  - number of turns
  - time
  - resistance**
- S.I. unit of induction is: **(2005)**
  - Tesla**
  - Henry
  - Watt
  - Weber
- The current produced by moving the loop of a wire across the magnetic field is called: **(2001)**
  - Electric Current
  - A.C. Current
  - D.C. Current
  - Induced Current**

**Mutual Induction & Self Induction:**

9. One Henry is equal to:  
 (a)  $Vs^{-1} A$  (b)  $VsA$  (c)  $Vs^{-1} A^{-1}$  (d)  $VsA^{-1}$  **(MP:25)**
10. Mutual induction is the working principle of:  
 (a) A.C generator **(b) transformer.** (c) copper loss (d) hysteresis loss **(2021)**
11. If the number of turns in a coil is doubled, its self-inductance will become:  
 (a) halved **(b) doubled** (c) threefold (d) fourfold **(2015, 11)**
12. The phenomenon of producing e.m.f. in the coil due to the change of current in the coil itself is called:  
 (a) Mutual induction **(b) Self-induction** (c) Magnetic flux (d) None of the above **(2001)**

**A.C Generator:**

13. In an alternating current generator electrical energy is obtained by the conversion of:  
 (a) wind energy (b) Mechanical energy (c) Solar energy (d) Chemical energy **(2025)**
14. This device converts electrical energy into mechanical energy:  
 (a) Generator (b) transformer **(c) electric motor** (d) transistor **(2016)**
15. When an A.C. generator is converted into a D.C. generator, slip ring is replaced by:  
 (a) a dynamo **(b) a split ring** (c) a field coil (d) an inductor **(2014, 13, 09)**
16. The device used to convert A.C. into D.C. is called:  
 (a) Amplifier (b) L.E.D. **(c) Commutator** (d) Rectifier **(2009)**

**Transformer:**

17. Working principle of transformer is:  
 (a) Mutual Induction (b) Self Induction (c) Charge (d) Energy **(MP:25)**
18. A transformer steps up or steps down:  
 (a) Energy (b) AC only (c) DC only (d) both AC & DC **(2022)**
19. The practical application of the phenomenon of mutual inductance is:  
 (a) A.C. generator. **(b) Transformer** (c) Rectifier (d) Dynamo **(2018,13, 12, 09)**
20. The core of a transformer is laminated to reduce the loss of Energy caused by:  
 (a) NA  $\phi$  [Flux leakage] **(b) Eddy current** (c) Heating (d) All of these **(2021, 11)**
21. Transformers are used in circuits containing.  
 (a) d.c alone **(b) a.c alone** (c) both a.c and d.c. (d) non-inductive winding **(2010)**
22. In a conventional transformer:  
 (a) the current moves from primary to secondary windings without any change.  
**(b) EMF is induced in the secondary by the changing magnetic flux**  
 (c) none of the above  
 (d) the heat is transferred from primary to secondary. **(2008)**
23. In stepdown transformer:  
 (a)  $N_s > N_p$  **(b)  $N_s < N_p$**  (c)  $N_s = N_p$  (d)  $N_s \approx N_p$  **(2021, 03)**
24. Transformer works on:  
 (a) Ohm's Law (b) Self-Induction **(c) Mutual Induction** (d) Gauss's Law **(2002E)**



**SECTION- B: SHORT QUESTIONS****Faraday's Laws, Lenz Law, Motional e.m.f:**

1. What is Induced e.m.f.? State the Faraday's Law and Lenz's Law of electromagnetic Induction. **(2002E)**
2. Define Electromagnetic Induction and Induced e.m.f. State Faraday's Laws of electromagnetic induction. Also state Lenz's law for induced e.m.f. How is it related to electromagnetic induction? **(99, 93)**
3. What is Electromagnetic Induction? Describe Faraday's Laws and Lenz's Law of Electromagnetic Induction. Show that Lenz's Law is in accordance with the Law of Conservation of energy. **(1996)**

**Mutual Induction & Self Induction:**

4. State Faraday's laws of Electromagnetic induction. Explain Mutual induction and derive an expression for mutual inductance. **(2016)**
5. Define coefficient of mutual induction. With the help of labeled diagram, derive the expression for the mutual induction of a pair of coils placed very close to one another. **(2001, 98, 96)**

**Energy Stored in Current Carrying Coil:**

6. How energy is stored in a current carrying coil. Derive the formula for energy produced in self-induction.

**Motional e.m.f:**

7. What is the name of the EMF induced in a straight conductor moving with uniform velocity in a magnetic field? Derive its mathematical expression. **(2025)**
8. Define Motional e.m.f. and derive the relevant formula. **(2014)**
9. In what way is the motional emf, produced in a conductor placed in a magnetic field, translated into the relevant mathematical relation? **(2012)**
10. What is motional e.m.f.? Derive an expression for it. **(2007, 03, 97, 93)**

**A.C Generator:**

11. Describe the working of an AC generator. **(MP:25)**
12. Describe the principle of operation of an AC generator. **(2024)**

**Transformer:**

13. What is a Transformer? On what principle does it work? Derive mathematical relationship between emf and number of turns in coils. **(2021)**

**NUMERICAL PROBLEMS:****Faraday's Laws:**

1. A coil having an area of cross-section  $0.05 \text{ m}^2$  and number of turns 100 is placed perpendicular to the magnetic field of induction  $0.08 \text{ weber/m}^2$ . How such EMF will be induced in it if the field is reduced to  $0.02$  in  $0.01 \text{ sec.}$ ? (Ans: 10 volts) **(1993)**

**Motional e.m.f:**

2. A conductor of length  $0.4 \text{ m}$  moves at a velocity of  $15 \text{ ms}^{-1}$  perpendicular to a magnetic field of  $0.3 \text{ T}$ . Calculate the motional EMF induced in the conductor. **(MP:25)**
3. An aero plane flying in a region, where the vertical component of earth's magnetic field is  $3.2 \times 10^{-4} \text{ T}$ . If the wings span of the aero plane is  $50 \text{ meters}$  and its velocity is  $360 \text{ km/hour}$ . Find the induced e.m.f between the tips of the wings. (Ans:  $1.6 \text{ V}$ ) **(2007)**

**Mutual Induction:**

4. An emf of  $45 \text{ mV}$  is induced in a coil of  $500$  turns. When the current in a neighboring coil change from  $10$  amperes to  $4$  amperes in  $0.2$  seconds.

- (a) What is the mutual inductance of the coils? (b) What is the rate of change of flux in the second coil?
5. An e.m.f. of 45 millivolts is induced in a coil of 500 turns. When the current in a neighboring coil change from 15 amps to 4 amps in 0.2 seconds, (a) what is the mutual inductance of the coils? (b) What is the rate of change of flux in the second coil? **(2015)**
6. A pair of adjacent coils has a mutual inductance of 850 mH. If the current in the primary coil changes from 0 to 20 A in 0.1 sec; what is the change in the magnetic flux in the secondary coil of 800 turns?  
(Ans: 0.02125 weber) **(2009)**
7. An emf of 45 milli volts is induced in a coil of 500 turns, when the current in a neighboring coil change from 10 amperes in 0.2 seconds. (i) What is the Mutual Inductance of the coils? (ii) What is the rate of change of flux in the second coil? (Ans:  $2.25 \times 10^{-3}$  H,  $90 \times 10^{-6}$  weber/sec) **(2002E)**
8. What will be the mutual inductance of two coils when the change of a current of 3 amperes in one coil produces the change of flux of  $6 \times 10^{-4}$  weber in the second coil having 2000 turns? (Ans: 0.4 H) **(2001)**

### Self-Induction:

9. An iron core solenoid with 400 turns has a cross-section area of  $4.0\text{cm}^2$ . A current of 2.0 ampere passing through it produces  $B = 0.2$  weber/ $\text{m}^2$ . What emf is produced in it if the current is turned off in 0.1 sec? What is its self-inductance? **(2023)**
10. An iron core solenoid with 600 turns has a cross-section area of  $2.0\text{cm}^2$ . A current of 4.0 ampere passing through it produces  $B = 0.4$  weber/ $\text{m}^2$ . What emf is produced in it, if the current is turned off in 0.2 second? What is its self-inductance? **(2016)**
11. An iron core solenoid with 500 turns has a cross section of  $5\text{cm}^2$ . A current of 2.3 ampere passing through it produces a flux of  $B=0.53$  Tesla. How large an e.m.f. is induced in it, if the current it turned off in 0.1 second? What is the self-inductance of solenoids? **(2014)**
12. The current in a coil of 500 turns is changed from 0 to 5.43 amps. Thereby producing a magnetic flux of  $8.52 \times 10^{-4}$  w. What is the Self-Inductance of the coil? (Ans: 0.07845 H) **(2004)**

### A.C Generator:

13. A coil of 400 turns in ac generator having an area of  $0.1\text{m}^2$  is rotating in a magnetic field of 50 T. in order to generate a maximum voltage of 220 volts, how fast is the coil to be rotated? Express your answer in revolutions / second. **(2019)**
14. An alternating current generator operates at 79Hz. The area of the coil is  $500\text{cm}^2$ . Calculate the number of turns in the coil when a magnetic field induction 0.06 tesla produces a maximum potential difference of 149 volts. **(2017)**
15. An A.C generator operates at 79 Hz. The area of the coil is  $500\text{cm}^2$ . Calculate the number of turns in the coil when the magnetic field of induction is  $0.06\text{w}/\text{m}^2$  produces a maximum potential difference of 149 volts. **(2013)**
16. An alternating current Generator operating at 50 Hz has a coil of 200 turns, while the coil has an area of  $120\text{cm}^2$ . Calculate the magnetic field intensity applied to rotate the coil to produce the maximum voltage of 240. (Ans: 0.318 T) **(2011)**
17. A 500-turn coil in an A.C Generator having an area of  $100\text{cm}^2$  rotates in a magnetic field of value 50 tesla. In order to generate 220 volts maximum, how fast is the coil to rotate? Express your answer in terms of the No of revolution per second. (Ans: 0.088 rad/sec, 0.014 rev/sec) **(1998)**

### Transformer:

18. A step-down transformer reduces 1100V to 220V. The power output is 12.5KW and the overall efficiency of the transformer is 90%. The primary winding has 1000 turns. How many turns do the secondary have? What is power input? What is the current in each coil? (Ans: 200 turns, 13888 W, 12.61 A, 56.81 A) **(2008)**

19. A step-down transformer having 4000 turns in primary is used to convert 4400 V to 220 V. The efficiency of the transformer is 90% and 9 KW output is required. Determine the input power, the number of turns in the secondary coil and current in the primary & secondary coil. (Ans: 10,000W, 200 turns, 2.27A, 40.9A) (2005)
20. A transformer has 1000 turns in a primary coil. If the input voltage of the transformer is 200 volts, what would be the number of turns of the secondary coil to obtain an output of 6.0 volts? (Ans: 30 turns) (1999)

### SECTION- C: DETAILED QUESTIONS:

#### *Faraday's Laws:*

1. Define Electromagnetic Induction and Induced e.m.f. State Faraday's Laws of electromagnetic induction. Also state Lenz's law for induced e.m.f. How is it related to electromagnetic induction? (99, 93)
2. What is Electromagnetic Induction? Describe Faraday's Laws and Lenz's Law of Electromagnetic Induction. Show that Lenz's Law is in accordance with the Law of Conservation of energy. (1996)

#### *Mutual Induction & Self Induction:*

3. State Faraday's laws of Electromagnetic induction. Explain Mutual induction and derive an expression for mutual inductance. (2016, 14)
4. State Faraday's Laws of electromagnetic induction. Explain the phenomenon of mutual induction and derive an expression for mutual induction of two coils, also give its units. (2012, 07)
5. Define Self Induction & Self-Inductance and obtain an expression for the Mutual Inductance of two coils. (2009, 00, 96)

#### *A.C Generator:*

6. What is AC generator? Describe its working with graphical representation: Also derive relevant expression for E.M.F induce in A.C generator.
7. Draw a labeled diagram of an A.C. generator and derive expression for the alternating voltage produced. (Working is not required) (2009)
8. Describe with a neat diagram the working of Alternating Current Generator. (2004)

#### *Transformer:*

9. Define Self or Mutual Induction. Describe working of Transformer and derive relative expression. (MP:25)
10. What is Transformer? On what principle does it work? With the help of diagrams, give its construction and working. Derive expressions for its efficiency and turns ratio. (2023)
11. What is a Transformer? Write its types with the help of a clear diagram, give its construction and working and derive the relevant expression. (2016)